

| Sr. no | Short Questions                                                                  |
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| 1      | Define artificial intelligence (AI) and its main goal.                           |
| 2      | What is a production system in AI?                                               |
| 3      | Explain the concept of state space in problem-solving.                           |
| 4      | What is heuristic search and why is it used in AI?                               |
| 5      | Differentiate between generate-and-test and hill climbing search.                |
| 6      | What are the characteristics of a well-defined problem in AI?                    |
| 7      | Define knowledge representation in AI and its importance.                        |
| 8      | What is the frame problem in knowledge representation?                           |
| 9      | What is procedural knowledge, and how does it differ from declarative knowledge? |
| 10     | Explain the concept of nonmonotonic reasoning.                                   |
| 11     | What is the role of Bayes' Theorem in AI?                                        |
| 12     | Define a Bayesian network and its components.                                    |
| 13     | What is Dempster-Shafer theory used for in AI?                                   |
| 14     | What is a neural network, and what is its main function in AI?                   |
| 15     | Explain the difference between symbolic AI and connectionist AI.                 |
| 16     | What is a recurrent neural network (RNN), and what is it used for?               |
| 17     | Describe the three levels of robot programming.                                  |
| 18     | What is the role of an activation function in a neural network?                  |

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| 19 | Explain the concept of rule-based systems in AI.                                                              |
| 20 | What is the difference between forward and backward reasoning in AI?                                          |
|    | Long Question                                                                                                 |
| 21 | Discuss the history and key milestones of artificial intelligence (AI).                                       |
| 22 | Describe the basic elements of AI, including search, knowledge, and abstraction.                              |
| 23 | Explain the different types of search algorithms in AI and their characteristics.                             |
| 24 | Compare and contrast best-first search with problem reduction methods.                                        |
| 25 | What is constraint satisfaction, and how is it applied in AI search problems?                                 |
| 26 | Discuss the different approaches to knowledge representation in AI.                                           |
| 27 | Explain how the Is-a relationship and instance relationships are represented in AI using predicate knowledge. |
| 28 | Describe the significance of logic programming in AI with an example.                                         |
| 29 | What are the key principles of nonmonotonic reasoning, and how do they differ from classical logic?           |
| 30 | Explain certainty factors and their role in rule-based systems.                                               |
| 31 | Describe the structure and functionality of a Bayesian network in AI.                                         |
| 32 | Compare the Dempster-Shafer theory to Bayesian networks in handling uncertainty.                              |
| 33 | Explain how Hopfield networks and recurrent networks differ in their application in AI.                       |
| 34 | Discuss the learning process in neural networks and the key challenges associated with it.                    |
| 35 | Describe how distributed representations work in connectionist models and their relevance in AI.              |
| 36 | Explain the fundamentals of robot kinematics and their importance in AI-driven robotics.                      |

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| 37 | Describe the dynamic analysis and forces involved in robotic movement.                         |
| 38 | Explain the key challenges peculiar to robot programming languages and how they are addressed. |
| 39 | Discuss how means-ends analysis can be applied in AI to solve complex problems.                |
| 40 | Describe symbolic reasoning and statistical reasoning in AI, and compare their applications.   |